ONE K-THEORY TO RULE THEM ALL

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Abstract. Depending on if you are speaking to an operator algebraist, a topologist, or an algebraic geometer, you will get a different definition of K-theory. These distinct mathematical constructions have the same general purpose—to act as algebraic invariants for more sophisticated (and often unwieldly) mathematical objects. We carry out the initial constructions of the various K-groups—operator, topological, and algebraic—and conduct some preliminary investigations on the extent to which these objects are in fact deserving of having the same name.

Contents

1. Introduction 1
2. Preliminary constructions 2
2.1. Operator K-theory 2
2.2. Topological K-theory 8
2.3. Algebraic K-theory 17
3. Relating the Various K-Theories 20
3.1. A brief note on uniqueness of functors 20
3.2. Relating operator K-theory and topological K-theory 21
3.3. Relating topological K-theory and algebraic K-theory 23
3.4. Relating operator K-theory and algebraic K-theory 26
3.5. Tying everything together 29
4. Conclusion 30
References 31

1. Introduction

One way that a student of mathematics is reminded that the subject they are learning is of human origins is the extent to which mathematical structures and theorems are given evocative names and notation. Indeed, whether it be a straightforward connection between notation and terminology, such as the symbol $\mathbf{N}$ denoting the natural numbers, or a more brute-forced attempt to evoke imagery, such as the proof of the nonexistence of a continuous non-vanishing tangent vector field on S^{2k} being dubbed the "Hairy Ball Theorem", mathematicians seem to relish the opportunity to introduce creative nomenclature into their work. While this practice often has the obvious benefit of making mathematical terms and symbols easier to remember, trouble arises from the fact that uniqueness of terminology seems to be a local, but not global, property of the landscape of mathematics.

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Thus, for example, Wikipedia cites the existence of over 20 mathematical objects, spanning a wide range of fields, which are defined as being "normal". Of course, this lack of creativity in naming conventions could be forgiven in cases where repetitions in terminology corresponded to genuine connections between the underlying mathematical concepts. For example, there is clear utility in using the term "isomorphism" to define the general construction of a structure-preserving bijection in a variety of mathematical fields. Alternatively, in the case of "normal" mathematics, there does not seem to exist any sensible mathematical thread tying together the notions of a normal subgroup to a normal topological space (and even if there was, it certainly would not make a detour through the lands of probability theory).

At first glance, another potential culprit of this mathematical is the subfield of research known as K-theory. As far as the author is aware, there are at least three areas of mathematics—operator algebras, algebraic topology, and algebraic geometry—whose practitioners would claim that the subject of K-theory lies naturally as a subfield of their area of expertise. To distinguish between what these three distinct factions of mathematicians mean by the term "K-theory", the subject has been subdivided into operator K-theory, topological K-theory, and algebraic K-theory. A reasonable question to ask would be how reasonable is it for these three branches of mathematics, belonging to distinct fields, be grouped together under the shared head of the name"K-theory". That is, does there exist some precise mathematical sense in which the various K-theories can be related to one another, thus proving that they are in fact simply different perspectives by which to view the same general mathematical phenomenon? In this expository article, we will outline the existence of a mathematical thread which does establish the close rigorous connections which exist between the various K-theories. This thread will travel from the land of algebraic topology, through the picturesque fields (or more precisely the rings) of module theory, and finally arrive in the familiar pastures of operator algebras.

2. Preliminary constructions

2.1. **Operator K-theory.** Operator K-theory is concerned with associating algebraic invariants to C^*-algebras (and more generally Banach algebras). As we have covered much of the constructions involving operator K-theory in class, our exposition of this material is terse. Relevant proofs can be found in chapters 2 and 3 of [7].

Notations 2.1. Throughout, let A denote a C^*-algebra with unit unless otherwise specified.

We want to associate a C^*-algebra to A in analogous way as to how we associate the C^*-algebra of bounded operators $B(H)$ to a finite-dimensional Hilbert space H.

Construction 2.2 (Matrix algebra)**.** Given $n \in \mathbf{N}$, define $M_n(A)$ to be the set of $n \times n$-matrices with entries in A. This is given a vector space structure with operations defined entry-wise and is a $*$-algebra under matrix multiplication and involution given by the conjugate transpose. To make $M_n(A)$ a C^*-algebra it remains to define a norm which respects the $*$-algebra structure and makes $M_n(A)$ complete. We do this as follows. Let $i : A \hookrightarrow B(H)$ be a $*$-homomorphism

embedding for some fixed Hilbert space H. Define the map

$$i_n : M_n(A) \longrightarrow B(H^n)$$
$$i_n(a_{ij}) : \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \longmapsto \begin{pmatrix} \sum_{j=1}^{n} i_n(a_{1j})(v_1) \\ \vdots \\ \sum_{j=1}^{n} i_n(a_{nj})(v_n) \end{pmatrix}.$$

We can then define a norm on $M_n(A)$ by $\|(a_{ij})\| := \|i_n(a_{ij})\|_{B(H^n)}$. This norm is independent of the monomorphism i_n and it makes $M_n(A)$ a C^*-algebra.

Definition 2.3. Given $n \in \mathbf{N}$, we call the C^*-algebra $M_n(A)$ defined above a **matrix algebra** associated to A.

Definition 2.4. A **projection** in A is an element $p \in A$ such that $p^2 = p = p^*$. We denote the set of projections in A by $\mathcal{P}(A)$.

Definition 2.5. A **unitary** in A is an element $u \in A$ such that $uu^* = 1 = u^*u$. We denote the set of unitaries in A by $\mathcal{U}(A)$.

We introduce the following equivalence relations on the class $\mathcal{P}(A)$ of projections of A.

Definition 2.6. Let $p, q \in \mathcal{P}(A)$.

1. We say p and q are **Murray-von Neumann equivalent**, denoted $p \sim q$, if there exists $v \in A$ such that $p = v^*v$ and $q = vv^*$.
2. We say p and q are **unitary equivalent**, denoted $p \sim_u q$, if there exists a unitary $u \in \mathcal{U}(A)$ such that $p = uqu^*$.
3. We say p and q are **homotopic**, denoted $p \sim_h q$, if there exists a continuous path of projections in A from p to q.

Definition 2.7. Given projections $p, q \in \mathcal{P}(A)$, we say p and q are **orthogonal**, denoted $p \perp q$, if $pq = 0 = qp$.

Addition on A descends to a binary operation on orthogonal elements of $\mathcal{P}(A)/\sim$.

Proposition 2.8. *Given $p, q, p', q' \in \mathcal{P}(A)$, if $p \perp q, p' \perp q', p \sim p'$, and $q \sim q'$, we have $p + q \sim p' + q'$.*

We have defined these equivalence relations on $\mathcal{P}(A)$ in order of increasing strength. Indeed, we have the following.

Proposition 2.9. *Let $p, q \in \mathcal{P}(A)$.*

1. *If $p \sim_h q$ then $p \sim_u q$.*
2. *If $p \sim_u q$ then $p \sim q$.*

It would be nice if these three seemingly distinct notions of equivalence were in fact equivalent, but unfortunately this is not the case for a general C^*-algebra A. If we pass from A to the matrix algebra associated to A however, we get the desired result.

Proposition 2.10. *Let $p, q \in \mathcal{P}(A)$.*

1. *If $p \sim q$, then*
$$\begin{pmatrix} p & 0 \\ 0 & 0 \end{pmatrix} \sim_u \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix}$$
in $\mathcal{P}(M_2(A))$.
2. *If $p \sim_u q$, then*
$$\begin{pmatrix} p & 0 \\ 0 & 0 \end{pmatrix} \sim_h \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix}$$
in $\mathcal{P}(M_2(A))$.

An important point to note is that, up to Murray-von Neumann equivalence, the projection
$$\begin{pmatrix} p & 0 \\ 0 & 0 \end{pmatrix}$$
in $M_2(A)$ does not depend on the location of the entry p. Indeed, this stems from the fact that Murray-von Neumann equivalence coincides with unitary equivalence in the matrix algebra setting.

Proposition 2.11. *Let $i, i' : A \rightrightarrows M_n(A)$ be the $*$-homomorphisms defined by*
$$i : p \longmapsto (a_{ij}) \text{ where } a_{ij} = p \cdot \delta_{i,j}$$
$$j : p \longmapsto (a'_{ij}) \text{ where } a'_{ij} = p \cdot \delta_{i',j'},$$
where $\delta_{i,j}$ denotes the Kronecker delta in the unital ring A. Then $i(p) \sim_u j(p)$ for each projection $p \in A$.

Proof. We prove the case for $i = j$ and $i' = j'$ since that is the only case we will need for the construction of the semigroup of projections of A. Let $p_1 \coloneqq i(p)$ and $p_2 \coloneqq j(p)$ We have $p' = qpq'$ where $q = (b_{ij})$ is given by $b_{ij} = \delta_{i',i}$ and $q' = (b_{ij})$ is given by $b_{ij} = \delta_{i,i'}$. Since $1^* = 1$, q' is the conjugate transpose of q, hence $p' = qpq^*$, i.e. $p' \sim_u q$, as required. □

We now have the necessary requisite background to construct the operator K-groups. Explicitly, we will define the K_0 group of the C^*-algebra A to be the universal enveloping group of an associated semigroup of projections

Construction 2.12 (The semigroup of projections)**.** Define $\mathcal{P}_n(A) \coloneqq \mathcal{P}(M_n(A))$ and furthermore define the direct limit
$$\mathcal{P}_\infty(A) \coloneqq \varinjlim_n \mathcal{P}_n(A)$$
under the monomorphisms
$$i : \mathcal{P}_n(A) \ni p \longmapsto \begin{pmatrix} p & 0 \\ 0 & 0 \end{pmatrix} \in \mathcal{P}_{n+1}(A).$$
We consider the set $\mathcal{P}_\infty(A)/\sim$ of Murray-von Neumann equivalence classes of projections in $\mathcal{P}_\infty(A)$. In particular, we define the binary operation $\oplus$ on $\mathcal{P}_\infty(A)/\sim$ as follows: given $[p], [q] \in \mathcal{P}_n(A)$, let $p' \in [p]$ and $q' \in [q]$ such that $p', q' \in M_n(A)$ and $p' \perp q'$ (this can always be done by shifting q' down the diagonal, and this procedure preserves the equivalence class by Proposition 2.9). Then we define
$$[p] \oplus [q] \coloneqq [p' + q'].$$
This operation is well-defined by Proposition 2.8, and furthermore it makes $\mathcal{P}_\infty(A)/\sim$ into a semigroup with identity $[0]$.

We are now ready to define the first operator K-group of A.

Definition 2.13. We define the $\mathbf{0^{th}}$ **K-group** of A, denoted $K_0(A)$, to be the universal enveloping group of the semigroup $\mathcal{P}_\infty(A)/\sim$.

As one would hope, the construction of the K_0 group associated to a C^*-algebra is functorial.

Proposition 2.14. *Given a morphism of semigroups $f : A \to B$, the universal property of the universal enveloping group of B defines a unique abelian group homomorphism from the universal enveloping group of A to the universal enveloping group of B. Furthermore, this construction extends the universal enveloping group to a functor $UEG : \mathsf{SGrp} \to \mathsf{Ab}$.*

Construction 2.15 (Extending K_0 to a functor)**.** Firstly, for each $n \in \mathbf{N}$, the matrix algebra construction $A \mapsto M_n(A)$ extends to an endofunctor on the category $\mathsf{C^*Alg}$ of C^*-algebras and $*$-homomorphisms, as follows: given a $*$-homomorphism $f : A \to B$, $f_n : M_n(A) \to M_n(B)$ is defined by evaluating f on $(a_{ij}) \in M_n(A)$ entrywise. This constructs a sequence of $*$-homomorphisms f_n, $n \in \mathbf{N}$, which commute with the embedding morphisms $i_n : M_n(A) \hookrightarrow M_{n+1}(A)$ and $j_n : M_n(B) \hookrightarrow M_{n+1}(B)$. Thus, in the unital case, the sequence of $*$-homomorphisms f_n descend to a unique semigroup homomorphism $f_* : \mathcal{P}_\infty(A) \to \mathcal{P}_\infty(B)$, i.e., the semigroup construction $A \mapsto \mathcal{P}_\infty(A)$ extends to a functor $\mathcal{P}_\infty^\sim$ from the full subcategory $\mathsf{C^*Alg}_1^{\text{full}}$ of $\mathsf{C^*Alg}$ consisting of unital C^*-algebras, to the category SGrp of semigroups. Finally, composing with the universal enveloping group functor $\mathsf{SGrp} \to \mathsf{Ab}$ gives an abelian group homomorphism $K_0(f) : K_0(A) \to K_0(B)$.

Proposition 2.16. *Let $f : A \to B$ be a $*$-homomorphism between unital C^*-algebras. The abelian group homomorphism $K_0(f) : K_0(A) \to K_0(B)$ defined above makes K_0 into a functor of the form $\mathsf{C^*Alg}_1 \to \mathsf{Ab}$.*

The K_0 functor satisfies has some useful structure-preserving properties. In particular it is split exact.

Proposition 2.17 (K_0 is split exact)**.** *Given a split exact sequence diagram in $\mathsf{C^*Alg}^{\text{full}}$, the image of this diagram under K_0 is split exact in Ab.*

2.1.1. *Operator K-groups of nonunital C^*-algebras.* Clearly the definition of the K_0-group of a C^*-algebra A defined in the previous section relies on the notion of unitary elements of A, hence on A having a unit. To extend K_0 along the inclusion functor $\mathsf{C^*Alg}_1^{\text{full}} \hookrightarrow \mathsf{C^*Alg}$, we first devise a construction by which we can functorially embed a nonunital C^*-algebra into a corresponding unital C^*-algebra. This construction is given as follows.

Definition 2.18. Suppose A is nonunital. We define the **unitization** $\widetilde{A}$ of A to be the C^*-algebra given as follows:

- the underlying set of $\widetilde{A}$ is $A \times \mathbf{C}$;
- vector space operations and the $*$-operation on $\widetilde{A}$ are defined coordinate wise;
- the norm $\|\cdot\|_{\widetilde{A}}$ on $\widetilde{A}$ is given by

$$\|(a,\lambda)\|_{\widetilde{A}} := \max(|\lambda|, \sup\{\|a'a\|_A : a' \in A, \|a'\|_A \leq 1\}).$$

It is not too difficult to verify that the above construction does define a legitimate C^*-algebra $\widetilde{A}$. In fact, even if A did have a unit, $\widetilde{A}$ would still be a C^*-algebra.

What follows are some facts concerning the unitization functor.

Proposition 2.19. *Let a C^*-algebra A, potentially without unit.*

1. *The $*$-monomorphism $i : A \hookrightarrow \widetilde{A}$ given by $a \mapsto (a, 0)$ embeds A as an ideal of $\widetilde{A}$, and furthermore $A/\widetilde{A} \cong \mathbf{C}$ as C^*-algebras.*
2. *There exists a split exact sequence*

$$0 \longrightarrow A \hookrightarrow \widetilde{A} \underset{\mu}{\overset{\pi}{\rightleftarrows}} \mathbf{C} \longrightarrow 0 \ ,$$

   where π is the projection map onto the quotient algebra $\widetilde{A}/A \cong \mathbf{C}$ and μ is defined by $\mu : \lambda \mapsto (0, \lambda)$.
3. *If A has a unit 1, then the map $p = (-1, 1)$ is projection in $\widetilde{A}$ and the map $f : A \oplus \mathbf{C} \to \widetilde{A}$ defined by $(a, \lambda) \mapsto (a - \lambda, \lambda)$ is a $*$-isomorphism between $A \oplus \mathbf{C}$ and $\widetilde{A}$.*

Proposition 2.20. *Given a $*$-homomorphism $f : A \to B$ of (possibly nonunital) C^*-algebras, the $*$-homomorphism $\widetilde{f} : \widetilde{A} \to \widetilde{B}$ defined by evaluating $(a, \lambda) \in \widetilde{A}$ by f on the first component and $1_{\mathbf{C}}$ on the second component makes the unitization construction into an endofunctor $\widetilde{(-)} : \mathsf{C^*Alg} \to \mathsf{C^*Alg}$ valued in the subcategory $\mathsf{C^*Alg}_1$ of unital C^*-algebras and unit-preserving $*$-homomorphisms.*

We are now ready to define the K_0-group of a C^*-algebra in the nonunital case.

Construction 2.21 (K_0 of a nonunital C^*-Algebra)**.** Let A be a nonunital C^*-algebra. Define the abelian group $K_0(A)$ by

$$K_0(A) := \ \ker(K_0(\widetilde{A}) \xrightarrow{K_0(\pi)} K_0(\mathbf{C})) \ ,$$

where π is the projection map defined in the split exact sequence in proposition 2.19.

Furthermore, if $f : A \to B$ is a $*$-homomorphism between non-unital C^*-algebras, by the universal property of the kernel of $K_0(\widetilde{B})$, there is a unique dashed morphism $K_0(f)$ such that the following commutative diagram (obtained through the functoriality of K_0 on $\mathsf{C^*Alg}_*$) commutes:

$$\begin{tikzcd}
K_0(A) \arrow[r] \arrow[d, dashed, "K_0(f)"'] & K_0(\widetilde{A}) \arrow[r, "K_0(\pi_A)"] \arrow[d, "K_0(\widetilde{f})"] & K_0(\mathbf{C}) \arrow[d, equal] \\
K_0(B) \arrow[r] & K_0(\widetilde{B}) \arrow[r, "K_0(\pi_B)"'] & K_0(\mathbf{C})
\end{tikzcd} \ .$$

For the construction above of non-unital K-theory to be a valid extension of the K_0 functors to $\mathsf{C^*Alg}$, it must agree with K_0 for the unital case. This is in fact the case.

Proposition 2.22. *Let A be a unital C^*-algebra. Then $K_0(A) = \ker(\widetilde{A} \xrightarrow{K_0(\pi)} K_0(\mathbf{C}))$. Furthermore, if $f : A \to B$ is a $*$-homomorphism of unital C^*-algebras, $K_0(f)$*

makes the following commutative diagram commute:

$$\begin{tikzcd} K_0(A) \arrow[r] \arrow[d, dashed, "K_0(f)"'] & K_0(\widetilde{A}) \arrow[r, "K_0(\pi_A)"] \arrow[d, "K_0(\widetilde{f})"] & K_0(\mathbf{C}) \arrow[d, equal] \\ K_0(B) \arrow[r] & K_0(\widetilde{B}) \arrow[r, "K_0(\pi_B)"'] & K_0(\mathbf{C}) \end{tikzcd} .$$

Proof. Since A is unital, then by exactness of the K_0 functor on $\mathsf{C^*Alg}$, $\ker(K_0(\pi)) = \mathrm{im}(K_0(\iota)) = K_0(A)$, where $\iota : A \hookrightarrow \widetilde{A}$ is the usual embedding.

This diagram is the image under K_0 of a diagram in $\mathsf{C^*Alg}_1$, with $f : A \to B$ the left-most arrow in the preimage diagram. Functoriality of K_0 implies $K_0(\varphi)$ makes the corresponding diagram in Ab commute, as required. □

Thus we have a valid extension of K_0 to a functor on $\mathsf{C^*Alg}$. Consequently we define the K_0-group of a general C^*-algebra to be its image under this K_0 functor.

Definition 2.23. Given an arbitrary C^*-algebra A, we define it $\mathbf{0^{th}}$ **K-group** $K_0(A)$ to be the image of A under the functor $K_0 : \mathsf{C^*Alg} \to \mathsf{Ab}$ defined above.

2.1.2. *Higher operator K-groups.* We can extend the above K-group construction given above to yield a sequence of functors K_n from the category of C^*-algebras into the category of abelian groups. The key idea behind the construction of the higher K-functors is that, for a given $n \in \mathbf{N}$, K_n is given as the composite of the K_0 functor pre-composed n times with a particular endofunctor on $\mathsf{C^*Alg}$. We define this endofunctor as follows.

Definition 2.24. The **suspension** SA of A is the C^*-algebra $C_0([0,1], A)$, i.e.

$$SA \coloneqq \{f \in C([0,1], A) : f(0) = f(1) = 0\}$$

with pointwise operations and sup norm.

Given a $*$-algebra homomorphism $f : A \to B$, sending f to the pushforward map $f_* : SA \to SB$ makes suspension of C^*-algebras into an endofunctor

$$S : \mathsf{C^*Alg} \longrightarrow \mathsf{C^*Alg},$$

which restricts to an endofunctor on $\mathsf{C^*Alg}_1^{\mathrm{full}}$ (thus we could define the higher K-groups on $\mathsf{C^*Alg}_1^{\mathrm{full}}$ only).

Definition 2.25. We define the $\boldsymbol{n^{th}}$**-K-group** of A, denoted $K_n(A)$, as the K_0 group of the n-th suspension of A, i.e.

$$K_n(A) \coloneqq K_0(S^n A).$$

This defines a functor $K_n : \mathsf{C^*Alg} \to \mathsf{Ab}$.

2.1.3. *Bott periodicity for operator K-theory.*

Theorem 2.26 (Bott Periodicity I)**.** *There exists a natural isomorphism*

$$K_0 \overset{\beta}{\underset{\cong}{\Rightarrow}} K_2 \ .$$

2.2. **Topological K-theory.** The K^0 group of a compact topological space X, as in the case of operator K-theory, arises as the universal enveloping group of a particular semigroup associated associated to X. There is, however some differences in terminology which highlight differences in the historical development of these theories.

Notation 2.27.

1. Throughout, we let $\mathbf{k}$ denote either $\mathbf{R}$ or $\mathbf{C}$.
2. We denote the category of $\mathbf{k}$-vector space by $\mathsf{Vect}_{\mathbf{k}}$.

Conventions 2.28.

1. Throughout, we assume all finite-dimensional $\mathbf{k}$-vector spaces we are dealing with are topologized with the usual topology induced by $\mathbf{k}^n$.
2. Since the corresponding K-theoretical results are considerably more well-behaved (e.g. to have the usual Bott-periodicity statement hold), throughout we assume X is an arbitrary fixed, compact Hausdorff topological space. All the definitions we make work just as well for non-compact spaces.
3. Given an abelian semigroup M, the model we use for the universal enveloping group of M is the standard one involving the quotient of the abelian semigroup $M \times M$ under the identification $(a, b) \sim (c, d)$ iff there exists $\lambda \in M$ such that
$$a + d + \lambda = b + c + \lambda.$$

Definition 2.29. An ***n*-dimensional *k*-vector bundle** over X is a collection $\{E_x\}_{x \in X}$ of k-vector spaces and a topology on the set $E \coloneqq \amalg_{x \in X} E_x$ so that the map

$$\begin{aligned} \pi : E &\longrightarrow X \\ E_x \ni e &\longmapsto x \end{aligned}$$

satisfies the following **local triviality condition**: each $x \in X$ has a neighborhood $U \ni x$ and a homeomorphism

$$\varphi_U : U \times k^n \xrightarrow{\cong} E|_U \coloneqq \pi^{-1}(U)$$

such that

1. $\pi(\varphi_U(x, v)) = x$ for all $(x, v) \in U \times \mathbf{k}^n$;
2. φ_U restricts to a linear isomorphism between $\{u\} \times \mathbf{k}^n$ and E_u for each $u \in U$.

We call such a U a **trivializing open set**, and φ_U a **trivialization**. We denote the vector bundle by $\pi : E \to X$ or just E (with the existence of a map π implied implicitly).

The local triviality condition implies that the dimension of fibers E_x over X is locally constant. In particular, the following definition makes sense.

Definition 2.30. Let E be a vector bundle over X.

1. We define the **rank function** of E to be the map
$$\begin{aligned} \mathrm{rank}_E : X &\longrightarrow \mathbf{N} \\ x &\longmapsto \dim(E_x). \end{aligned}$$

2. If X is connected, then rank_E is constant, and we define the unique integer $\mathrm{rank}(E) \in \mathrm{im}\,\mathrm{rank}_E$ to be the **rank** of E.
3. If X has basepoint x_0, we define its **rank**, also denoted by $\mathrm{rank}(E)$, by
$$\mathrm{rank}(E) \coloneqq \mathrm{rank}_E(x_0).$$

Remark 2.31. Clearly the two notions of rank agree in the case that (X, x_0) is a connected based space, and thus the notion of the rank of a connected or based vector bundle is well-defined.

Definition 2.32. A **section** of a vector bundle $\pi : E \to X$ is a continuous map $s : X \to E$ such that $\pi \circ s = 1_X$. We denote the set of sections of π by $\Gamma(X, E)$.

Upon coming across the definition of a mathematical object for the first time, perhaps the most practical question one may ask themselves is what is the simplest concrete object which satisfies said definition. Such objects are often known as "trivial" cases of their respective definition, and the case of the vector bundle is no different.

Definition 2.33. Given a space X and a finite-dimensional $\mathbf{k}$-vector space V, we call the map $\underline{V}_X : X \times V \to X$ given by projection onto the first coordinate (with $X \times V$ having the product topology and $V_x = \{x\} \times V$ the evident $\mathbf{k}$-vector space structure) a **trivial vector bundle** over X.

A reasonable next step in developing the theory of a newly defined mathematical object is to define a notion of morphisms between such objects.

Definition 2.34. If $\pi : E \to X$ and $\pi' : E' \to X$ are vector bundles over a space X, then a map $f : E \to E'$ is a **vector bundle map** from π to π' if it is linear on each fiber, i.e. $f_x : E_x \to E'_x$ is a linear map. We say that π and π' are **isomorphic** if there exists vector bundle maps $f : E \leftrightarrows E' : g$ such that $g \circ f = 1_E$ and $f \circ g = 1_{E'}$.

Remark 2.35. Up to isomorphism the trivial vector bundles over X are indexed by $\mathbf{N}$: Given $n \in \mathbf{N}$, $\underline{V}_X \cong \underline{\mathbf{k}^n}_X$ for every $\mathbf{k}$-vector space V of dimension n.

Now that we have objects and morphisms between those objects, we can consider the category consisting of such objects and morphisms.

Definition 2.36. Given a space X, we denote by $\mathsf{Vect}_{\mathbf{k}}(X)$ the **category of k-vector bundles** and vector bundle morphisms between them.

2.2.1. *Pullback of vector bundles.* We want to define a functor on the category Top of topological spaces which sends a space X to the category $\mathsf{Vect}_{\mathbf{k}}(X)$ of $\mathbf{k}$-vector bundles over itself. In order to do this we need a proper notion of how a map of spaces induces a functor between vector bundle categories. The key construction this theory rests on is that of the *pullback bundle*. Firstly, we define the category our vector bundle functor will be valued in.

Definition 2.37. We denote by Cat the category whose objects are locally-small categories and whose morphisms are functors between them.

Construction 2.38 (Pullbacks)**.** Let $f : Y \to X$ be a continuous map and let $\pi : E \to X$ a vector bundle over X. Denoting by f^*E the pullback $Y \times_X E$, define the map $f^*\pi : f^*E \to Y$ by projection onto the first coordinate. It follows that $f^*\pi$ is a vector bundle over Y.

Definition 2.39. Given a vector bundle E over X and a continuous map $f : Y \to X$, we call the vector bundle f^*E defined above the **pullback** of E along f.

The pullback construction is functorial.

Proposition 2.40. *Let $g : E \to F$ be a map between vector bundles E and F over X and let $f : Y \to X$ be a continuous map. The map f^*g given by*

$$\begin{aligned} f^*g : f^*E &\longrightarrow f^*F \\ (y, e) &\longmapsto (y, g(e)) \end{aligned}$$

is a vector bundle morphism. Furthermore, since the existence of f^*g follows from the universal property of the pullback f^*E, this extends the vector bundle construction $X \mapsto \mathsf{Vect}_{\mathbf{k}}(X)$ into a contravariant functor $\mathsf{Vect}_{\mathbf{k}} : \mathsf{Top}^{\mathrm{op}} \to \mathsf{Cat}$.

2.2.2. *Whitney sums.* The category $\mathsf{Vect}_{\mathbf{k}}(X)$ has some interesting properties stemming from the fact that we can combine vector bundles together to construct new ones. We call this newly constructed vector bundle the *Whitney sum* (or *direct sum*). The Whitney sum is nothing more than the[1] product in the category $\mathsf{Vect}_{\mathbf{k}}(X)$. We will see shortly that since $\mathsf{Vect}_{\mathbf{k}}(X)$ is additive, the Whitney sum also serves as coproducts in $\mathsf{Vect}_{\mathbf{k}}(X)$. Our coverage of the Whitney sum construction is terse. Proofs we do not include can be found in section I.4 of [6].

Construction 2.41 (Whitney sums)**.** Let $\pi : E \to X$ and $\pi' : E' \to X$ be $\mathbf{k}$-vector bundles. Denote the pullback $E \times_X E'$ by $E \oplus E'$ and consider the map $\pi \oplus \pi' : E \oplus E' \to X$ which commutes with the pullback diagram. It follows that $(E \oplus E')_x = E_x \times E'_x$ has the natural structure of a $\mathbf{k}$-vector space with dimension $\dim(E \oplus E') = \dim E_x + \dim E'_x$.

We claim $\pi \oplus \pi'$ is a $\mathbf{k}$-vector bundle with rank $r_{\pi\oplus\pi'} = r_\pi + r_{\pi'}$. It remains to check that the triviality condition is satisfied. Given $x \in X$, denote $\mathrm{rank}_\pi(x)$ by n and $\mathrm{rank}_{\pi'}(x)$ by n'. There exists trivializing open neighborhoods U and U' of x and x' and trivializations $\varphi|_U : U \times \mathbf{k}^n \to E_U$ and $\varphi|_{U'} : U' \times \mathbf{k}^{n'} \to E_{U'}$. Defining $V := U \cap U'$ and

$$\begin{aligned} \varphi|_V : V \times \mathbf{k}^{n+n'} &\longrightarrow E \oplus E' \\ (v, (w, w')) &\longmapsto (\varphi|_U(v, w), \varphi|_{U'}(v, w')), \end{aligned}$$

it follows that V is an open neighborhood of x such that given $(v, (w, w')) \in V \times \mathbf{k}^{n+n'}$, we have $\pi(\varphi|_U(v, w)) = v = \pi(\varphi|_{U'}(v, w'))$ and thus $(\pi \oplus \pi') \circ \varphi|_V$ corresponds to projection onto the first coordinate, as required. The fact that $\mathrm{rank}_{\pi\oplus\pi'} = \mathrm{rank}_\pi \oplus \mathrm{rank}_{\pi'}$ follows immediately from the definition of the rank function and the fact that $(E \oplus E')_x = E_x \times E'_x$.

Definition 2.42. Given vector bundles $\pi : E \to X$ and $\pi' : E' \to X$, the vector bundle $\pi \oplus \pi'$ defined above is called the **Whitney sum** (or the **direct sum**) of π and π'. We often denote $\pi \oplus \pi'$ as $E \oplus E'$.

Compactness of X yields the following somewhat surprising result that all vector bundles, up to an appropriate Whitney sum, are trivial.

[1]We use the definite article "the" to refer to the product as (co)limit objects are defined up to unique isomorphism in their respective category, which is the proper notion of uniqueness in category theory.

Theorem 2.43. *Given a vector bundle E over X, there exists a vector bundle E' such that $E \oplus E'$ is trivial.*

Definition 2.44. An **additive category** A is a category which is enriched over Ab, has a zero object, and contains all finite biproducts (i.e. all finite (co)products).

Lemma 2.45. $\mathsf{Vect}_{\mathbf{k}}(X)$ *is additive with zero object* $\underline{\mathbf{k}}^0{}_X$.

Proof. Given bundles $\pi : E \to X$ and $\pi' : F \to X$ $\mathrm{Hom}_{\mathsf{Vect}_{\mathbf{k}}(X)}(E, F)$. □

The Whitney sum construction implies $\mathsf{Vect}_{\mathbf{k}}(X)$ has finite products.

Theorem 2.46. *The category* $\mathsf{Vect}_{\mathbf{k}}(X)$ *of vector bundles over X is an additive category.*

2.2.3. *Constructing K^0.*

Notation 2.47. Given a category C, we denote by $\mathsf{C}_{\cong}$ the quotient set of $\mathrm{ob}(\mathsf{C})$ by the equivalence relation generated by identifying isomorphic objects in C. Given $x \in \mathsf{C}$, we denote the equivalence class x belonging to in $\mathsf{C}_{\cong}$ by $[x]$.

We demonstrate how to associate an abelian monoid to $\mathsf{Vect}_{\mathbf{k}}(X)$ in the general setting of an arbitrary additive category.

Lemma 2.48. *Let A be an additive category. Claim $\mathsf{A}_{\cong}$ with binary operation*

$$[x] + [y] \coloneqq [x \oplus y] \qquad \textit{for all } [x], [y] \in \mathsf{A}_{\cong}$$

is an abelian monoid.

Proof. First we must verify this binary operation is well-defined. This follows from the fact that that the biproduct operator $(- \oplus -) : \mathsf{A} \times \mathsf{A} \to \mathsf{A}$ is functorial and hence preserves isomorphisms, hence $x \oplus y \cong x' \oplus y'$, as required. The abelian monoid structure on $\mathsf{A}_{\cong}$ follows from the fact that $(A, \oplus, 0)$ is a symmetric monoidal category, thus we get the necessary isomorphisms from the corresponding coherence diagrams. □

Definition 2.49. Given an additive category A, we call the universal enveloping group of the abelian monoid $\mathsf{A}_{\cong}$ the **Grothendieck group** of A, and we denote it by $K(\mathsf{A})$.

Remark 2.50 (Functoriality of K). An additive functor $F : \mathsf{A} \to \mathsf{B}$ induces a monoid homomorphism $\mathsf{A}_{\cong} \to \mathsf{B}_{\cong}$ in the obvious way, and by functoriality of the universal enveloping group, this defines an abelian group homomorphism $\varphi_* : K(\mathsf{A}) \to K(\mathsf{B})$.

The notation used for the Grothendieck group leaves little secret as to what the definition of the topological K^0-group of a space should be.

Definition 2.51. Given a compact space X, we define the $\mathbf{0^{th}}$ **K-group** of X, denoted $K^0(X)$, to be the Grothendieck group of $\mathsf{Vect}_{\mathbf{k}}(X)$.

We recall some standard facts from universal enveloping groups in the context of Grothendieck groups of additive categories to conclude some preliminary results about the K^0-group.

Proposition 2.52. *Given an additive category* A*, denote by* $[E]$ *the element in the Grothendieck group* $K(\mathsf{A})$ *corresponding to the isomorphism class of* $E \in \mathsf{A}$ *under the monoid embedding* $\mathsf{A}_{\cong} \hookrightarrow K(A)$*. Then every element of* $K(\mathsf{A})$ *can be represented in the form* $[E]-[F]$ *for some* $E, F \in \mathsf{A}$*, and furthermore* $[E]=[E']$ *iff there exists* $F \in \mathsf{A}$ *such that* $[E]+[F]=[E']+[F]$*.*

Definition 2.53. Given vector bundles E and F over X, we call the element $[E]-[F]$ in $K^0(X)$ a **virtual vector bundle** over X.

Thus by the proposition above, $K^0(X)$ denotes the group of virtual vector bundles over X.

Proposition 2.43 allows us to conclude that every element of $K^0(X)$ can be expressed by a virtual vector bundle of the following form.

Proposition 2.54. *Every element of* $K^0(X)$ *can be expressed as a difference* $[\underline{\mathbf{k}}^n]-[E']$ *for some* $n \in \mathbf{N}$ *and some* $E' \in \mathsf{Vect}_{\mathbf{k}}(X)$*.*

We know from remark 2.23 that the Grothendieck group construction defines a covariant functor on the category of additive categories and additive functors. In contrast, the K^0 group of a topological space defines a contravariant functor from the category CHaus of compact Hausdorff topological spaces.

Proposition 2.55. *Given a map* $f : X \to Y$ *in* Top*, there exists a map* $f^* : K^0(Y) \to K^0(X)$ *given by*

$$f^* : [E]-[F] \longmapsto [f^*(E)]-[f^*(F)],$$

which defines a functor $K^0 : (\mathsf{Top})^{\mathrm{op}} \to \mathsf{Ab}$*.*

2.2.4. *Reduced K-theory.* Throughout this section let X have basepoint x_0. Then the map

$$\begin{aligned} \mathrm{rk}_{x_0} : \mathsf{Vect}_{\mathbf{k}}(X, x_0)_{\cong} &\longrightarrow \mathbf{Z} \\ [E] &\longmapsto \mathrm{rank}_{x_0}(E) \end{aligned}$$

is an abelian monoid homomorphism. It follows by the universal property of the Grothendieck group construction that rk_{x_0} extends to an abelian group homomorphism

$$\mathrm{rk}_{x_0} : K^0(X, x_0) \longrightarrow \mathbf{Z}.$$

Furthermore, we have a split exact sequence

$$0 \longrightarrow \ker(\mathrm{rk}_{x_0}) \longrightarrow K^0(X) \underset{\mu}{\overset{\mathrm{rk}_{x_0}}{\rightleftarrows}} \mathbf{Z} \longrightarrow 0 \ ,$$

where μ is defined by $n \mapsto \mathrm{sgn}(n)\underline{\mathbf{k}^{|n|}}_X$.

Definition 2.56. Given a based compact Hausdorff space (X, x_0) **reduced $\mathbf{0}^{th}$ K-group** $\widetilde{K}^0(X, x_0)$ by

$$\widetilde{K}^0(X, x_0) := \ \ker(K^0(X) \xrightarrow{\mathrm{rk}_{x_0}} \mathbf{Z}) \ .$$

We sometimes omit the basepoint in the notion of the reduced K-group $\widetilde{K}^0(X)$.

We can equivalently define the K^0-group $\widetilde{K}^0(X)$ of X to by

$$\widetilde{K}^0(X) = \ker(K^0(X) \xrightarrow{i^*} K^0(\{x_0\}) \ ,$$

where $i : \{x_0\} \hookrightarrow X$ is the inclusion map.

By the above split exact sequence, if X is a based space, we have a splitting

$$K^0(X) \cong \widetilde{K}^0(X) \oplus \mathbf{Z}.$$

As in the unreduced case, $\widetilde{K}^0(X)$ can be thought of as a functor in X

Proposition 2.57. *Given a based map $f : X \to Y$ between based compact Hausdorff spaces (X, x_0) and (Y, y_0), there exists a unique abelian group homomorphism $f^* : \widetilde{K}^0(Y) \to \widetilde{K}^0(X)$, defined to be the restriction of f^* to $\widetilde{K}^0(Y)$, which makes the following diagram commute:*

$$\begin{tikzcd}
0 \arrow[r] & \widetilde{K}^0(Y) \arrow[r] \arrow[d, dashed, "\widetilde{f}^*"'] & K^0(X) \arrow[r, "\mathrm{rk}_{x_0}"] \arrow[d, "f^*"] & \mathbf{Z} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \widetilde{K}^0(X) \arrow[r] & K^0(X) \arrow[r, "\mathrm{rk}_{y_0}"'] & \mathbf{Z} \arrow[r] & 0
\end{tikzcd}.$$

Furthermore this construction extends the reduced K^0 group of a pointed compact Hausdorff space to a functor $\widetilde{K}^0 : \mathsf{CHaus}_*^{\mathrm{op}} \to \mathsf{Ab}$.

Proof. This follows immediately from the universal property of the kernel $\widetilde{K}^0(X)$. □

2.2.5. *Ring structure on Topological K-theory.* We can upgrade $K^0(X)$ to a ring with multiplication induced by the tensor product operation on vector bundles. In particular, $\mathsf{Vect}_{\mathbf{k}}(X)_{\cong}$ becomes a commutative semiring under the operation

$$[E] \otimes [F] \coloneqq [E \otimes F] \qquad \text{for all } [E], [F] \in \mathsf{Vect}_{\mathbf{k}}(X)_{\cong}.$$

Then the operation $- \otimes -$ is extended to the K_0 group of X in the obvious way: given $[E] - [F], [E'] - [F'] \in K_0(X)$, we have

$$([E] - [F]) \otimes ([E'] - [F']) \coloneqq [E] \otimes [F] - [E] \otimes [F'] - [F] \otimes [E'] + [F] \otimes [F'].$$

One can verify that this operation makes $K_0(X)$ into a commutative ring with unit $\mathbf{1} \coloneqq \underline{\mathbf{k}^1}_X$.

Notation 2.58. We denote $\sum_{i=1}^{n} \mathbf{1}$ in $K_0(X)$ by $\boldsymbol{n}$.

It follows that if (X, x_0) is based, the rank function $\mathrm{rk}_{x_0} : \mathsf{Vect}_{\mathbf{k}}(X)_{\cong} \to \mathbf{Z}$ becomes a ring homomorphism, and furthermore its kernel $\widetilde{K}^0(X, x_0)$ is an ideal, hence also a ring but not necessarily with unit.

Finally, we define the following important ring homomorphism.

Definition 2.59. Given spaces $X, Y \in \mathsf{CHaus}$, we define the **external product** of X and Y to be the map

$$\begin{aligned} \mu : K^0(X) \otimes K^0(Y) &\longrightarrow K(X \times Y) \\ a \otimes b &\longmapsto \pi_1^*(a)\pi_1^*(b), \end{aligned}$$

where $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ are the usual projection maps. We denote the image of $a \otimes b$ under μ by $a * b$.

2.2.6. *K-theory as a cohomology theory.* From this point on we only work with complex K-theory, i.e. we will assume throughout that $\mathbf{k} = \mathbf{C}$, i.e. we are dealing with $\mathbf{C}$-vector bundles over our spaces. Our desire to expand the K^0 functor to an extraordinary cohomology theory K^n motivates the following definitions.

Definition 2.60. Let $n < 0$.

1. We inductively define the $\boldsymbol{n^{th}}$ **K-group** of X, denoted by $K^n(X)$, by
$$K^n(X) \coloneqq K^{n+1}(\Sigma X),$$
where ΣX denotes the unreduced suspension of X.
2. Given a closed subspace $A \subset X$, we define the $\boldsymbol{n^{th}}$ **K-group** of the pair of spaces (X, A) by $K^n(X, A) \coloneqq K^n(X/A)$.

The definitions of $\widetilde{K}^{-n}$ and $\widetilde{K}(X, A)$ in the reduced setting are analogous.

We have that the K^0 functor, and hence the K^{-n} functor for $n \in \mathbf{N}$, descends to a functor on the homotopy category $\mathsf{Ho}(\mathsf{CHaus})$.

Proposition 2.61. *Given homotopic maps $f, g : X \rightrightarrows Y$, $f^* = g^*$ as maps $K^0(Y) \to K^0(X)$.*

Next we would like to prove that K^{-n} satisfies the exactness axiom, i.e., we would like to construct the long exact sequence of the K-groups of a pair. Constructing such a long exact sequence is significantly easier for reduced K-groups, and thus this is the setting we will work in. Throughout, we assume X has basepoint $* \in X$.

For brevity we make the following abuse of notation.

Notation 2.62. From now on, given a map $f : X \to Y$ of based spaces in CHaus_*, we denote the associated abelian group homomorphism $\widetilde{K}^0(Y) \to \widetilde{K}^0(X)$ induced by the functor $\widetilde{K}^0$ in the same way as in the unreduced case, i.e. by f^*.

The following result is a significant step in the construction of the long exact sequence of a pair.

Proposition 2.63. *Given a closed subset $A \subset X$ and a basepoint $* \in A \subset X$, we have that the sequence*
$$A \xrightarrow{\ i\ } X \xrightarrow{\ q\ } X/A$$
induces an exact sequence of abelian groups
$$\widetilde{K}^0(A) \xleftarrow{\ i^*\ } \widetilde{K}^0(X) \xleftarrow{\ q^*\ } \widetilde{K}^0(X/A)\ .$$

We can extend this exact sequence to the left as follows. Given a pair (X, A) with $A \subset X$ compact hausdorff spaces, we have the following sequence of inclusions of pushouts in CHaus:

$$\begin{array}{ccccccccccc} A & \overset{i}{\hookrightarrow} & X & \overset{i}{\hookrightarrow} & X \cup CA & \overset{i}{\hookrightarrow} & (X \cup CA) \cup CX & \overset{i}{\hookrightarrow} & ((X \cup CA) \cup CX) \cup C(X \cup CA) & \overset{i}{\hookrightarrow} & \dots, \\ & & & & \downarrow q & & \downarrow q & & \downarrow q & & \\ & & & & X/A & & \Sigma A & & \Sigma X & & \end{array}$$

where C denotes the cone endofunctor on CHaus and the vertical maps are the quotient maps collapsing the most recently attached cone. The fact that the cone of a space is contractible is very useful when paired with the following lemma.

Lemma 2.64. *If $(X, A, *)$ is a based pair of compact Hausdorff spaces with $A \subset X$ closed, the quotient map $q : X \to X/A$ induces an isomorphism $\widetilde{K}^0(X/A) \cong \widetilde{K}^0(X)$.*

The proof is obvious in the case where (X, A) is sufficiently nice, i.e. the inclusion $i : A \hookrightarrow X$ is a cofibration, since then the quotient map $q : X \to X/A$ would be a homotopy equivalence (and thus would be sent to a group isomorphism by the functoriality of $\widetilde{K}^0$). The proof for a general compact Hausdorff pair (X, A) is more involved, and is contained in section 2.2 of [5].

We combine the above proposition and lemma to conclude that the above sequence of pushouts induces the following one-sided long-exact sequence of $\widetilde{K}$-groups:

$$\widetilde{K}^0(A) \xleftarrow{i^*} \widetilde{K}^0(X) \xleftarrow{q^*} \widetilde{K}^0(X/A) \xleftarrow{\partial} K^{-1}(A) \xleftarrow{i^*} K^{-1}(X) \xleftarrow{q^*} \widetilde{K}^{-1}(X/A) \leftarrow \dots,$$

where ∂ is the map $(qi)^* : \widetilde{K}^0(\Sigma A) \to \widetilde{K}^0(X \cup CA) \cong \widetilde{K}^0(X/A)$.

To extend this to a full long-exact sequence of a compact Hausdorff pair $(X, A, *)$, we will need the Bott periodicity theorem for topological K-theory. First, we construct a reduced version of the external product $\mu : a \otimes b \mapsto a * b$.

Construction 2.65 (The external product on reduced K-theory)**.** Given based compact Hausdorff spaces X and Y, the one-sided long exact sequence of the pair $(X \times Y, X \vee Y)$ yields a splitting

$$\widetilde{K}^0(X \times Y) \cong \widetilde{K}^0(X \wedge Y) \oplus \widetilde{K}^0(X) \oplus \widetilde{K}^0(Y).$$

We also have the following isomorphism of rings

$$\begin{aligned} K(X) \otimes K(Y) &\cong (\widetilde{K}^0(X) \oplus \mathbf{Z}) \otimes (\widetilde{K}^0(Y) \oplus \mathbf{Z}) \\ &\cong \widetilde{K}^0(X) \otimes \widetilde{K}(Y)^0 \oplus (\widetilde{K}(X) \otimes \mathbf{Z}) \oplus (\widetilde{K}(Y) \otimes \mathbf{Z}) \oplus (\mathbf{Z} \otimes \mathbf{Z}) \\ &\cong \widetilde{K}^0(X) \otimes \widetilde{K}^0(Y) \oplus \widetilde{K}^0(X) \oplus \widetilde{K}^0(Y) \oplus \mathbf{Z}. \end{aligned}$$

It follows, after some diagram chasing, that we can define a ring homomorphism $\widetilde{\mu} : \widetilde{K}^0(X) \otimes \widetilde{K}^0(Y) \to \widetilde{K}^0(X \wedge Y)$ as the restriction map of the unreduced external product given by the following diagram

$$\begin{tikzcd} K(X) \otimes K(Y) \arrow[d, "\mu"'] & \cong & (\widetilde{K}^0(X) \otimes \widetilde{K}^0(Y)) \arrow[d, dashed, "\widetilde{\mu}"'] & \oplus & \widetilde{K}^0(X) \oplus \widetilde{K}^0(Y) \arrow[d, equal] & \oplus & \mathbf{Z} \arrow[d, equal] \\ K^0(X \times Y) & \cong & \widetilde{K}(X \wedge Y) & \oplus & \widetilde{K}(X) \oplus \widetilde{K}(Y) & \oplus & \mathbf{Z} \end{tikzcd}.$$

Definition 2.66. Given X and Y based compact Hausdorff spaces, the ring homomorphism

$$\widetilde{\mu} : \widetilde{K}^0(X) \otimes \widetilde{K}^0(Y) \longrightarrow \widetilde{K}^0(X \wedge Y)$$

defined above is called the **reduced external product** of X and Y. Again we denote the image of $a \otimes b$ under $\widetilde{\mu}$ by $a * b$.

2.2.7. *Bott periodicity for topological K-theory.* There is a particular class of 1-dimensional vector bundles which are distinguished in topological K-theory.

Definition 2.67. The **canonical line bundle** over $\mathbf{C}P^n$ is the vector bundle $\pi : E \to \mathbf{C}P^n$ where

$$E \coloneqq \{(l, v) \in \mathbf{C}P^n \times \mathbf{C}^{n+1} : v \in l\}$$

and π is defined by projection onto the first coordinate.

Definition 2.68. The canonical line bundle over $\mathbf{C}P^1 \cong S^2$ is called the **Bott bundle**, and we denote its isomorphism class in $\mathsf{Vect}_{\mathbf{C}}(S^2)_{\cong}$ by H.

Proposition 2.69 (The canonical line bundle is nontrivial)**.** *$H \neq \mathbf{1}$.*

In fact, more is known about the structure of H as an element of the K-group $K^0(S^2)$.

Lemma 2.70. *In $K^0(S^2)$, $H^2 + 1 = 2H$, or equivalently $(H-1)^2 = 0$.*

Thus we have a ring homomorphism

$$\psi : \mathbf{Z}[H]/((H-1)^2) \to K^0(S^2).$$

Compactness of X gives us the following result, a proof of which is contained in section 2.1 of [5].

Theorem 2.71 (Fundamental Product Theorem)**.** *The ring homomorphism φ defined as the composite of*

$$K^0(X) \otimes \mathbf{Z}[H])/(H^2-1)^2 \xrightarrow{1_{K^0(X)} \otimes \psi} K^0(X) \otimes K^0(S^2) \xrightarrow{\mu} K^0(X \times S^2)$$

is an isomorphism.

Taking X to be a point yields the following:

Corollary 2.72. *The map $\mathbf{Z}[H]/(H-1)^2 \to K^0(S^2)$ is a ring isomorphism.*

Construction 2.73 (The Bott map)**.** We have that the n-fold reduced suspension $S^n \wedge X$ is obtained from the n-fold unreduced suspension $\Sigma^n X$ by collapsing the points $[x_0, t]$, with $t \in I^n$, to a single point. Since $\{[x_0, t] \in \Sigma^n X\} \cong D^n$, hence contractible, it follows by Lemma 2.64 that the quotient map $\Sigma^n X \to S^n \wedge X$ induces an isomorphism on $\widetilde{K}^0$-groups.

Precomposing this isomorphism, for the case $n = 2$, with the reduced external product μ yields a homomorphism

$$\begin{aligned} \widetilde{K}^0(S^2) \otimes \widetilde{K}^0(X) &\longrightarrow \widetilde{K}^0(\Sigma^2 X) \\ s \otimes x &\longmapsto s * x \end{aligned}$$

Recall $K^0(S^2) = \mathbf{Z}[H]/((H-1)^2)$ and thus $\widetilde{K}^0(S^2)$ is the free abelian group generated by $H - 1$. Thus, using the map we defined above, we get a homomorphism

$$\begin{aligned} \beta : \widetilde{K}^0(X) &\longrightarrow \widetilde{K}^0(\Sigma^2 X) \\ x &\longmapsto (H-1) * x. \end{aligned}$$

Definition 2.74. We call the homomorphism $\beta : K^0(X) \to K^0(S^2 X)$ defined above the **Bott map** of X.

Finally we can state and prove the Bott periodicity theorem.

Theorem 2.75 (Bott Periodicity II)**.** *The Bott map β of X is an isomorphism.*

Proof. By the construction above, we have that β is given as the composition

$$\widetilde{K}^0(X) \xrightarrow{x \mapsto (H-1) \otimes x} \widetilde{K}^0(S^2) \otimes \widetilde{K}^0(X) \xrightarrow{\widetilde{\mu}} K^0(X),$$

where the first map is an isomorphism since $\widetilde{K}^0(S^2) \cong \mathbf{Z}$ with generator $H - 1$, and the second map is an isomorphism by the Fundamental Product Theorem 2.71. $\square$

Corollary 2.76. *The Bott map induces a natural isomorphism between K^0 and K^{-2}.*

Thus, by Bott periodicity, up to isomorphism there are two distinct complex K-theory cohomology functors.

Definition 2.77. Given $n \in \mathbf{Z}$, we define the $\boldsymbol{n^{th}}$ **K-theory functor**

$$\widetilde{K}^n : \mathsf{CHaus}_*^{\mathrm{op}} \longrightarrow \mathsf{Ab}$$

by

$$\widetilde{K}^n := \begin{cases} \widetilde{K}^0, & n \equiv 0 \mod 2 \\ \widetilde{K}^{-1}, & n \equiv -1 \mod 2 \end{cases}.$$

Bott periodicity along with the definition of the reduced K-theory functors generalize to the unreduced case. This makes the job of verifying that K^n satisfies the axioms of a cohomology theory significantly easier.In particular the one-sided long exact sequence defined above collapse to a six-term periodic exact sequence.

2.3. **Algebraic K-theory.** Algebraic K-theory, at least in the classical sense of Grothendieck, associated a K_0-group to a ring R in an analogous way to how topological K-theory associates $K_0(X)$ to a topological space X. The key difference between the two constructions is that in the algebraic setting, the additive category we are working with is that of finitely-generated projective modules over R. We begin with some conventions and basic definitions.

Conventions 2.78.

1. We fix an arbitrary ring R with unit $1 \neq 0$ satisfying the invariant basis property (i.e. such that $R^n \ncong R^m$ as R-modules for $n \neq m$).
2. The word R-module will always mean right R-module.

If R is a field, an R-module is called a vector space, and we know from linear algebra that vector spaces satisfy a variety of nice properties. Perhaps the nicest property satisfied by vector spaces is that they all have a minimal spanning set, called a *basis*. The cardinality of a given basis in fact does not depend on the particular choice of its elements, and thus it defines an invariant of the underlying vector space, known as the *dimension* (*or rank*) of the vector space.

Modules over an arbitrary ring R does not satisfy such nice properties in general. Since vector spaces are very well-behaved mathematical objects precisely because of these properties, exactly under what conditions an R-module does behave like a vector space is an important area of study.

Definitions 2.79. Let M be an R-module.

1. A **basis** for M is a subset $E := \{e_i\}_{i \in I}$ of M such that every element of M can be expressed uniquely as a finite R-linear combination of elements in E.
2. We say that M is **free** if it has a basis.
3. If M has a fixed ordered basis E, we call M a **based free module** and define the **rank** of M to be the cardinality of E.

Example 2.80 (Canonical free module)**.** Given $n \in \mathbf{N}$, the R-module R^n with the usual basis $E = (e_i)_{i=1}^n$, where e_i has 1 in the ith entry and 0 elsewhere, is a free module. In fact, every free R-module of rank n is isomorphic to R^n, with module isomorphism generated by a bijection between bases.

Definition 2.81. An R-module P is said to be **projective** if it satisfies the **projective lifting property**: for every surjection $s : M \to N$ of R-modules and every homomorphism $g : P \to N$, there exists a homomorphism $f : P \to M$ such that $sf = g$.

There is an equivalent characterization of projective modules in terms direct summands of free modules (proof on page 8 of [6]).

Lemma 2.82. *An R-module P is projective iff there exists an R-module Q such that $P \oplus Q$ is free.*

Definition 2.83. Let $\mathsf{Mod}(R)$ denote the category of finitely generated R-modules (and R-module homomorphisms).

1. We define $\mathsf{Free}(R)$ to be the full subcategory of finitely generated free R-modules in $\mathsf{Mod}(R)$.
2. We define $\mathsf{Proj}(R)$ to be the full subcategory of projective R-modules in $\mathsf{Mod}(R)$.

2.3.1. Proj *is functorial.* Throughout this section we let S denote a fixed ring with 1.

Construction 2.84 (Functors between module categories)**.** Let $f : R \to S$ be a ring homomorphism.

Given an S-module M, we can define an R-module M_R with the same underlying set as M and R-scalar multiplication given by

$$m \cdot r = mf(r) \qquad \text{for each } m \in M \text{ and each } r \in R.$$

It follows that given an S-module homomorphism $g : M \to N$, as a map between sets g can naturally be considered an R-module homomorphism $g_R : M_R \to N_R$ since given $m \in M$ and $r \in R$, we have

$$\begin{aligned} g_R(mr) &= g_R(mf(r)) \\ &= g(m)f(r) \\ &= g_R(m)r. \end{aligned}$$

Thus this construction defines a functor

$$\rho_f : \mathsf{Mod}(S) \longrightarrow \mathsf{Mod}(R).$$

Conversely, given an R-module M, we can define an S-module M^S with underlying set $M^S = M \otimes_R S$, where S is considered to be a left R-module with R-scalar multiplication given by $r \cdot s = f(r)s$. The S-scalar multiplication on M^S is then defined by $(m \otimes s) \cdot s' \coloneqq m \otimes (ss')$.

Given an R-module homomorphism $g : M \to N$, the R-module homomorphism $g \otimes 1_S$ is, as a map between sets, naturally an S-homomorphism $g^S : M^S \to N^S$. Indeed, given $m \otimes s \in M^S$ and $s' \in S$, we have

$$\begin{aligned} g^S((m \otimes s)s') &= g^S(m \otimes (ss')) \\ &= g(m) \otimes (ss') \\ &= (g(m) \otimes s)s' \\ &= g^S(m \otimes s)s'. \end{aligned}$$

Thus this construction also defines a functor

$$\varepsilon_f : \mathsf{Mod}(R) \longrightarrow \mathsf{Mod}(S).$$

Definition 2.85. ,

1. We call the functor $\rho_f : \mathsf{Mod}(S) \to \mathsf{Mod}(R)$ defined above the **restriction of scalars along f**.
2. We call the functor $\varepsilon_f : \mathsf{Mod}(R) \to \mathsf{Mod}(S)$ define above the **extension of scalars along f**.

Theorem 2.86. *We have an adjunction $\varepsilon_f \dashv \rho_f$.*

Proposition 2.87. *Given a ring homomorphism $f : R \to S$ and a projective R-module P, the S-module P^S is projective.*

Proof. We want to show P^S satisfies the projective lifting property. Let $s : M \to N$ be a surjective S-module homomorphism and let $g : P^S \to M$ be an arbitrary homomorphism. Since ρ_f preserves the underlying map of sets, it follows that $s_R : M_R \to N_R$ is a surjective R-module homomorphism, hence by the fact that P is projective, there exists an R-module homomorphism $h : P \to N_R$ such that $s_R h = g_R$. Applying ε_f gives that $sh^S = g$, as required. □

Corollary 2.88 (Proj is functorial)**.** *The mapping*

$$\begin{aligned} \mathrm{Hom}_{\mathsf{Rng}_1}(R, S) &\longrightarrow \mathrm{Fun}(\mathsf{Proj}(R), \mathsf{Proj}(S)) \\ f &\longmapsto (\varepsilon_f)|_{\mathsf{Proj}(R)} \end{aligned}$$

extends the map $R \mapsto \mathsf{Proj}(R)$ into a functor $\mathsf{Proj} : \mathsf{Rng}_1 \to \mathsf{Cat}$.

Lemma 2.89. *Given projective R-modules P, Q, $P \oplus Q$ is projective.*

Proof. Let $r : M \to N$ be a surjective homomorphism and $g : P \oplus Q \to M$ be another homomorphism. By the projective lifting property applied to both P, Q, there exist homomorphisms $f_1 : P \to M$ and $f_2 : Q \to M$ such that $rf_k = gi_k$ for $k \in \{1, 2\}$, where $i_1 : P \hookrightarrow P \oplus Q$ and $i_2 : Q \hookrightarrow P \oplus Q$ are the homomorphism given the coproduct structure of $P \oplus Q$. It follows by the universal property of the coproduct that there exists a homomorphism $f : P \oplus Q \to M$ such that $fi_k = f_k$. Consequently, $rfi_k = rf_k = gi_k$ for $k \in \{1, 2\}$ and thus by universal property of the coproduct, $rf = g$, as required. □

Since $\mathsf{Mod}(R)$ is an additive category with zero object the trivial R-module $M = \{0\}$, the above lemma implies that $\mathsf{Prod}(R)$ inherits the additive structure from $\mathsf{Mod}(R)$.

Proposition 2.90. $\mathsf{Proj}(R)$ *is an additive category.*

Following the same construction of the topological K^0-group, we are immediately able to define the K_0-group of a ring R.

Definition 2.91. The **0th K-group** of R, denoted by $K_0(R)$, is defined as the Grothendieck group of the additive category $\mathsf{Proj}(R)$.

3. Relating the Various K-Theories

3.1. **A brief note on uniqueness of functors.** In category theory, the notion of objects (be they (co)limits, functors, natural transformations, etc.) being literally equal in the formalism of set theory is much too restrictive. In general one can at most expect an object to be unique only up to up unique isomorphism, and even uniqueness of isomorphisms cannot be relied upon as objects become less concrete (objects vs functors vs natural transformations, ...). In light of this, throughout this section wherever we state that a certain diagram of functors *commute*, we always mean implicitly that the relevant composites are isomorphic up to natural isomorphism (if not uniquely so), rather than being literally equal as maps of sets. We formalize this notion of commutativity of functors in the language of *2-categories*.

3.1.1. *The language of 2-categories.* The intuition behind a 2-category is that they are designed to account for higher-order structure between mathematical objects which are not captured at the level of ordinary morphisms between those objects (in the respective category). This higher-level structure is a general phenomenon found throughout category theory. For example, consider the fundamental groupoid $\Pi(X)$ of a space X, i.e. the category whose objects are the points of X and whose morphisms $[q] \in \mathrm{Hom}_{\Pi(X)}(x, y)$ are homotopy classes of paths $q : x \to y$. This standard category, what we call a *1-category*, misses out on a great deal of the higher-order topological structure of the space X. In particular, it does not encode the fact that distinct homotopic paths $p, p' \in [q]$ can be homotopic in *different ways*. This lies in the fact that $\Pi(X)$ is really the homotopy category $\mathsf{Ho}(\Pi_1(X))$ of the category $\Pi_1(X)$, whose objects are points in X and whose morphisms are paths in X between points. Put another way, if we consider homotopies of paths $H : p \simeq p'$ as paths of paths, i.e. as paths in the space $\{\gamma \in \mathsf{Top}(I, X) : \gamma(0) = x, \gamma(1) = y\}$, then it would be very reasonable to consider homotopy classes of such paths. In this way a homotopy class of a path of paths, i.e. a homotopy class $[H]$ of a homotopy $H : p \simeq p'$ between paths p and p', can be considered as a *morphism of morphisms* in $\Pi_1(X)$, also known as a *2-morphism*. Given paths $p, q \in \mathrm{Hom}_{\Pi_1}(x, y)$, we thus have a set $\mathrm{Hom}(p, q)$ of 2-morphisms between p and q in X, namely the set of homotopy classes of homotopies in X, i.e.

$$\mathrm{Hom}(p, q) \coloneqq \{[H] : H : p \simeq q \text{ in } X\}.$$

Thus, if we really wanted to consider all of the topological information of X, we would not pass to the homotopy category $\Pi(X)$ immediately, since only considering homotopy classes of paths in X collapses all higher-dimensional data down to a point (namely, all homotopies between homotopic paths, that is 2-morphisms, are identified in the homotopy category). Instead we would simultaneously consider the 1-category theoretic information of $\Pi_1(X)$, along with the 2-morphisms of $\Pi_1(X)$, i.e. sets of morphisms $\mathrm{Hom}(p, q)$ between morphisms $p, q \in \mathrm{mor}(\Pi_1(X))$. This new category-theoretic object, call it $\Pi_2(X)$, is an example of a 2-category. We formalize this example with the following definitions. This material can be found in section 3.3 of [4]

Clearly, cartesian product of categories makes Cat into a symmetric monoidal category with unit the category $\mathbf{1}$ with a single object and no non-identity morphisms.

Definition 3.1. A category $\mathfrak{C}$ which is enriched over Cat is called a **2-category**. Given objects x and y in $\mathfrak{C}$, the objects and morphisms of the category $\mathrm{Hom}_{\mathfrak{C}}(x, y)$ are called **1-morphisms** and **2-morphisms** from x to y respectively.

Prototypical Example 3.2 (Cat)**.** The category Cat of all locally-small categories and functors between them is naturally a 2-category, where given categories C and D, $\mathrm{Hom}_{\mathsf{Cat}}(\mathsf{C}, \mathsf{D})$ is the functor category $\mathrm{Fun}(\mathsf{C}, \mathsf{D})$, whose objects are functors between C and D and whose morphisms (i.e. the 2-morphisms in Cat) are natural transformations between them.

With the formalism of 2-categories we have established, we can revise our notion of commutativity of functors to account for the nuances we outlined at the beginning of this section regarding equality in a category-theoretic setting.

Definition 3.3. A diagram in a 2-category is said to be **2**-commutative, if its 1-morphisms commute up to given 2-isomorphisms and these 2-isomorphisms commute in the induced diagram taking 1-morphisms and their compositions as vertices.

Thus in the case of Cat, a diagram of functors is 2-commutative only if the composite functors that make up that diagram are naturally isomorphic by natural isomorphisms which commute (in the relevant functor categories). For example, we say the following diagram in Cat,

$$\begin{tikzcd} \mathsf{C} \arrow[r, "F"] \arrow[dr, "H"'] \arrow[dr, bend right, "L"'] & \mathsf{D} \arrow[d, "G"] \\ & E, \end{tikzcd}$$

is 2-commutative if there exists natural isomorphisms $\eta : GF \Rightarrow H$, $\varepsilon : H \Rightarrow L$, and $\nu : GH \Rightarrow L$ such that the following diagram commutes in $\mathrm{Fun}(\mathsf{C}, \mathsf{E})$:

$$\begin{tikzcd} GF \arrow[r, "\eta"] \arrow[dr, "\nu"'] & H \arrow[d, "\varepsilon"] \\ & L \end{tikzcd} .$$

For our purposes, we consider 2-commutativity to be the "right" notion of commutativity for diagrams in Cat.

Convention 3.4. Throughout this paper, when we say a "diagram of functors commutes", we implicitly mean the relevant diagram of functors is *2-commutative*.

3.2. **Relating operator K-theory and topological K-theory.** There are very clear similarities in the construction of the operator K-groups of a C^*-algebra A and the topological K-groups of a topological space X. Both $K_0(A)$ and $K_0(X)$ are defined as the universal enveloping group of some monoid associated to the underlying object and the higher K-groups $K^n(A)$ and $K^{-n}(X)$ are defined to be the K_0 groups of the n-th suspension (in the appropriate sense) of A and X respectively. Indeed the similarities in their constructions lead to a shared phenomenon characterizing the higher K-groups in both the operator-algebraic and the topological setting. Explicitly, the *Bott periodicity theorem* states that both the higher K-groups of a unital C^*-algebra and of a topological space are 8-fold periodic in the real setting and 2-fold periodic in the complex setting. We expand on the details of these ideas below.

To simplify things we will only consider the case of Bott periodicity in the complex setting, i.e. we will work with complex C^*-algebras and complex vector bundles over X.

In order to state the Gelfand-Naimark theorem we need to recall the notion of a spectrum of a C^*-algebra.

Definition 3.5. The **spectrum** $\widetilde{A}$ of a commutative C^*-algebra is defined as the set of non-zero $*$-homomorphisms from A into $\mathbf{C}$, topologized as a subspace the dual space A^* of A with the weak-* topology. Elements of $\widetilde{A}$ are called **characters** on A.

Lemma 3.6. *The spectrum $\widetilde{A}$ of a commutative C^*-algebra A is a locally compact Hausdorff space. Furthermore if A is unital then $\widetilde{A}$ is compact.*

It follows that given a unital $*$-homomorphism $f : A \to B$ between commutative, unital C^*-algebras A and B, the pullback map $f^* : \widetilde{B} \to \widetilde{A}$ extends the spectrum construction on $\mathsf{CC}^*\mathsf{Alg}$ into a contravariant functor

$$\widetilde{(-)} : \mathsf{CC}^{*\mathrm{op}} \longrightarrow \mathsf{CHaus}.$$

Notations 3.7.

1. We denote by CHaus the category of compact Hausdorff spaces (and continuous maps).
2. We denote by $\mathsf{CC}^*\mathsf{Alg}_1$ the category of commutative C^*-algebras and unit-preserving $*$-homomorphisms.

Construction 3.8 (Gelfand representation)**.** Given a commutative C^*-algebra A and an element $a \in A$, denote by ev_a the function $\widetilde{A} \to \mathbf{C}$ given by evaluating elements of the spectrum of A at a. We have that ev_a is continuous and vanishes at infinity, and furthermore that the map

$$\begin{aligned} \gamma : A &\longrightarrow C_0(\widetilde{A}) \\ a &\mapsto \mathrm{ev}_a \end{aligned} ‘$$

is a unit-preserving $*$-homomorphism.

Definition 3.9. Given a commutative C^*-algebra A, we call the $*$-homomorphism γ defined above the **Gelfand transform** of A.

We are now ready to state the commutative Gelfand-Naimark theorem.

Theorem 3.10 (Commutative Gelfand-Naimark)**.** *Let A be a commutative C^*-algebra. The Gelfand transform $\gamma : A \to C_0(\widetilde{A})$ is an isometric $*$-isomorphism.*

In the case where A is unital and hence $\widetilde{A}$ is compact, we have the following contravariant equivalence of categories implied by Gelfand-Naimark, which serves as the bridge between the two seemingly distinct branches of K-theory.

Corollary 3.11. *There exists a duality of categories between CHaus and $\mathsf{CC}^*\mathsf{Alg}_1$, given by the pair of functors*

$$\mathsf{CHaus}^{\mathrm{op}} \underset{\widetilde{(-)}}{\overset{C_0(-)}{\rightleftarrows}} \mathsf{CC}^*\mathsf{Alg}_1 \ .$$

Lemma 3.12. *There exists a natural isomorphism between the composite contravariant functors $SC_0 : \mathsf{CHaus}^{\mathrm{op}} \to \mathsf{CC}^*\mathsf{Alg}_1$ and $C_0\Sigma : \mathsf{CHaus} \rightrightarrows (\mathsf{CC}^*\mathsf{Alg}_1)^{\mathrm{op}}$.*

Theorem 3.13. *The complex K-theory cohomology functors factor through the category of commutative unital C^*-algebras and $*$-homomorphisms. Explicitly, given $n \in \mathbf{N}$, we have the following commutative diagram:*

$$\begin{tikzcd} \mathsf{CHaus}^{\mathrm{op}} \arrow[rr, "K^n"] \arrow[dr, "C_0"'] & & \mathsf{Ab} \\ & \mathsf{CC^*Alg}_1 \arrow[ur, "K_n"'] & \end{tikzcd}.$$

We hold off the proof of this theorem until we have established some formal connections between topological K-theory and algebraic K-theory, and between algebraic K-theory and operator K-theory.

3.3. **Relating topological K-theory and algebraic K-theory.** The striking similarities in the construction of algebraic K-theory and topological K-theory should lead one to suspect the existence of procedure for moving between the two constructions in some reasonable setting. Indeed, Jean Serre in the late 50s (in the case of algebraic varieties and algebraic K-theory) followed by Richard Swan a few years later (in the case of topological spaces which we will be using) showed this to be the case. We omit many technical proofs used in establishing this fundamental results, all of which can be found in I.6. of [6].

First we need to construct a suitable functor for transporting between the topological data of vector bundles over a compact space and the algebraic data of finitely-generated projective modules over a ring.

Construction 3.14 (The section functor)**.** Consider the ring $C(X, \mathbf{k})$ of continuous functions on X with values in $\mathbf{k}$. Given a k-vector bundle $\pi : E \to X$, the set $\Gamma(X, E)$ of continuous sections of π is a $C(X, \mathbf{k})$-module under the following operation: given $\lambda \in C(X, \mathbf{k})$ and $s \in \Gamma(X, E)$, we define the section

$$\begin{aligned} s \cdot \lambda : X &\longrightarrow E \\ x &\longrightarrow \lambda(x)s(x). \end{aligned}$$

Since a vector bundle map $f : E \to F$ descends to a module homomorphism $f_* : \Gamma(X, E) \to \Gamma(X, F)$ via post-composition by f, the construction associating a $C(X, \mathbf{k})$-module $\Gamma(X, E)$ to a vector bundle E extends to a functor

$$\Gamma : \mathsf{Vect}_{\mathbf{k}}(X) \longrightarrow \mathsf{Mod}(C(X, \mathbf{k})).$$

Lemma 3.15. *There exists $C(X, \mathbf{k})$-module isomorphisms*

1. $\Gamma(X, \underline{\mathbf{k}^n}_X) \cong C(X, \mathbf{k})^n$*;*
2. $\Gamma(X, E \oplus E') \cong \Gamma(X, E) \oplus \Gamma(X, E')$ *for each* $E, E' \in \mathsf{Vect}_{\mathbf{k}}(X)$*.*

Proof. 1. A section $s \in \Gamma(X, \underline{\mathbf{k}^n}_X)$ is characterized as a product of maps $s = s_1 \times \ldots s_n \times 1_X$, where $s_i \in C(X, \mathbf{k})$ (here we are implicitly precomposing the product on the right with the diagonal map $\Delta : X \to X^{n+1}$). Hence the map

$$\begin{aligned} f : \Gamma(X, \underline{\mathbf{k}}_X) &\longrightarrow C(X, \mathbf{k}) \\ s &\longmapsto (s_1, \ldots, s_n) \end{aligned}$$

clearly gives the desired module isomorphism.

2. Denote the vector bundle maps associated to E and E' by π and π' respectively. By the universal property of the pullback $E \oplus E'$, a section $s : X \to E \oplus E'$ is characterized by the pair of maps $\pi_1 s$ and $\pi_2 s$, where π_i is the projection map onto the i-th coordinate, such $\pi\pi_1 s = 1_X = \pi'\pi_2 s$. Thus $s_1 \coloneqq \pi_1 s$ and $s_2 \coloneqq \pi_2 s$ are sections of E and E' respectively. Consequently the map

$$\begin{aligned} f : \Gamma(X, E \oplus E') &\longrightarrow \Gamma(X, E) \oplus \Gamma(X, E') : \\ s &\longmapsto (s_1, s_2) \end{aligned}$$

clearly gives the desired module isomorphism, with the inverse homomorphism also defined by the universal property of the pullback.

□

The above proposition along with Theorem 2.28 allows us to conclude the following information about the section functor Γ.

Proposition 3.16.

1. *The section functor* $\Gamma : \mathsf{Vect}_{\mathbf{k}}(X) \to \mathsf{Mod}(C(X, \mathbf{k}))$ *takes values in the full subcategory* $\mathsf{Proj}(C(X, \mathbf{k}))$ *of projective* $C(X, \boldsymbol{k})$*-modules.*
2. Γ *is an additive functor.*

We can now state the major theorem relating topological and algebraic K-theory.

Theorem 3.17 (Serre-Swan)**.** *The section functor* Γ *induces an equivalence of categories* $\mathsf{Vect}_{\mathbf{k}}(X) \simeq \mathsf{Proj}(C(X, \mathbf{k}))$.

To prove such a theorem, we need to develop some additional categorical machinery.

Notation 3.18. Throughout, we let A denote a fixed additive category.

Definition 3.19. We say that A is **pseudo-abelian** if for every object $x \in \mathsf{A}$ and idempotent morphism $p \in \mathrm{Hom}_{\mathsf{A}}(x, x)$, the kernel of p exists.

There exists a universal procedure for embedding A into a pseudo-abelian category.

Theorem 3.20. *There exists a pseudo-abelian category* $\widetilde{\mathsf{A}}$ *and a fully faithful additive functor* $\varphi : \mathsf{A} \to \widetilde{\mathsf{A}}$ *satisfying the following universal property: For every pair of a pseudo-abelian category* B *and an additive functor* $\psi : \mathsf{A} \to \mathsf{B}$*, there exists an additive functor* $\widetilde{\psi} : \widetilde{A} \to \mathsf{B}$ *which is unique up to isomorphism such that the following diagram commutes up to isomorphism:*

$$\begin{tikzcd} \mathsf{A} \arrow[rr, "\psi"] \arrow[dr, "\varphi"'] & & \mathsf{B} \\ & \widetilde{\mathsf{A}} \arrow[ur, dashed, "\widetilde{\psi}"'] & \end{tikzcd} .$$

Moreover, the pair $(\widetilde{A}, \varphi)$ *is unique up to equivalence of categories.*

Definition 3.21. We call $\widetilde{\mathsf{A}}$ given in the above theorem the **pseudo-abelian category** associated to A.

Theorem 3.22. *Let* B *be a pseudo-abelian category and* $\psi : \mathsf{A} \to \mathsf{B}$ *be a fully faithful additive functor such that every object of* B *is a direct factor of an object in the image of* ψ *(i.e. given* $b \in \mathrm{ob}(\mathsf{B})$*, there exists* $a \in \mathrm{ob}(\mathsf{B})$ *and an idempotent morphism* $p : \psi(a) \to \psi(a)$ *in* B *such that* $b \cong \ker(p)$*). Then the functor* $\widetilde{\psi} : \widetilde{\mathsf{A}} \to \mathsf{B}$ *given by the universal property of* $\widetilde{A}$ *is an equivalence of categories.*

Theorems 3.16 and 3.17 are very powerful category-theoretic tools. Indeed, the following three results, one of which being a proof of Serre-Swan, all follow almost immediately from these two theorems.

Notation 3.23. We denote by $\mathsf{Vect}_{\mathbf{k}}^{\mathrm{triv}}(X)$ the full subcategory of $\mathsf{Vect}_{\mathbf{k}}(X)$ consisting of all trivial bundles.

Compactness of X yields the following result.

Theorem 3.24. *There exists an equivalence of categories between the universal pseudo-abelian category of* $\mathsf{Vect}_{\mathbf{k}}^{\mathrm{triv}}(X)$ *and* $\mathsf{Vect}_{\mathbf{k}}(X)$*.*

Theorem 3.25. *Let* R *be an arbitrary ring with unit. There exists an equivalence of categories between the pseudo-abelian category* $\widetilde{\mathsf{Free}(R)}$ *associated to* $\mathsf{Free}(R)$ *and* $\mathsf{Proj}(R)$*.*

We almost have everything we need to prove Serre-Swan. All that remains is a result which describes morphisms between trivial vector bundles.

Construction 3.26. Let $f : \underline{V}_X \to \underline{V'}_X$ be a morphism of trivial vector bundles. The commutative diagram

$$\begin{tikzcd} \underline{V}_X \arrow[rr, "f"] \arrow[dr] & & \underline{V'}_X \arrow[dl] \\ & X & \end{tikzcd}$$

implies that for each $x \in X$, f restricts to a linear map $f_x : \{x\} \times V \to \{x\} \times V'$.

Define the map

$$\begin{aligned} \check{f} : X &\longrightarrow \mathrm{Hom}_{\mathsf{Vect}_{\mathbf{k}}}(V, V') \\ x &\longmapsto f_x. \end{aligned}$$

Conversely, given a continuous map $h : X \to \mathrm{Hom}_{\mathsf{Vect}_{\mathbf{k}}}(V, V')$ (with the natural vector space topology induced by $\mathbf{k}^{\dim V \cdot \dim V'}$) define the map

$$\begin{aligned} \widehat{h} : \underline{V}_X &\longrightarrow \underline{V'}_X \\ \widehat{h} : (x, v) &\longmapsto (x, h(x)v). \end{aligned}$$

Theorem 3.27. *Let* $f : \underline{V}_X \to \underline{V'}_X$ *be a morphism of trivial vector bundles, and* $h : X \to \mathrm{Hom}_{\mathsf{Vect}_{\mathbf{k}}}(V, V')$ *be a continuous map.*

1. *The map* $\check{f} : X \to \mathrm{Hom}_{\mathsf{Vect}_{\mathbf{k}}}(V, V')$ *defined above is continuous.*
2. *The map* $\widehat{h} : \underline{V}_X \to \underline{V'}$ *defined above is a morphism of vector bundles.*
3. *The constructions* $\check{(-)}$ *and* $\widehat{(-)}$ *are mutual inverses, i.e.* $\widehat{\check{f}} = f$ *and* $\check{\widehat{h}} = h$*.*

,

Proof of theorem 3.17. The section functor induces a functor

$$\Gamma' : \mathsf{Vect}_{\mathbf{k}}^{\mathrm{triv}}(X) \to \mathsf{Free}(C(X,\mathbf{k})).$$

We claim Γ' is an equivalence of categories. Since $C(X,\mathbf{k})^n \cong \Gamma'(\underline{\mathbf{k}^n}_X)$, Γ' is essentially surjective. By Lemma 3.15, we fix bases $\{s_1,\dots,s_n\}$ for $\Gamma(X,\underline{\mathbf{k}}^n)$, where s_i is defined by $s_i : x \mapsto (e_i, x)$ (here $e_i \in \mathbf{k}^n$ is the standard i-th basis vector. It follows that given $\underline{k^m}_X$ and a morphism of vector bundles $f : \underline{k^n}_X \to \underline{k^m}_X$, $\Gamma'(f)$ is given by an $m \times n$-matrix $M = (a_{ij})$, where $a_{ij} \in C(X,\mathbf{k})$. We can equivalently consider M to be a continuous map

$$\begin{aligned} M : X &\longrightarrow M_{m,n}(\mathbf{k}) \\ x &\longmapsto (a_{ij}(x)). \end{aligned}$$

It follows that M coincides with the map $\widehat{f} : X \to \mathrm{Hom}_{\mathsf{Vect}_{\mathbf{k}}}(k^n, k^m)$. Furthermore, theorem 3.27 implies that Γ' is fully faithful, hence an equivalence of categories.

Let ψ denote the composite functor

$$\mathsf{Vect}_{\mathbf{k}}^{\mathrm{triv}}(X) \xrightarrow{\Gamma'} \mathsf{Free}(C(X,\mathbf{k})) \hookrightarrow \mathsf{Proj}(C(X,\mathbf{k})) \ .$$

Then we have the following commutative diagram of functors

$$\begin{tikzcd} \mathsf{Vect}_{\mathbf{k}}^{\mathrm{triv}}(X) \arrow[r, hook] \arrow[d, "\psi"'] & \mathsf{Vect}_{\mathbf{k}}(X) \arrow[dl, "\Gamma"] \\ \mathsf{Proj}(C(X,\mathbf{k})) & \end{tikzcd} .$$

By the universal property of the pseudo-abelian category $\mathsf{Vect}_{\mathbf{k}}(X)$, it follows that Γ is naturally isomorphic to $\widetilde{\psi}$. By theorem 3.22, Γ is an equivalence of categories, as required. □

The following corollary immediately from Serre-Swan, and makes explicit the connection between topological and algebraic K-theory.

Corollary 3.28. *We have the following commutative diagram of functors:*

$$\begin{tikzcd} \mathsf{Top}^{\mathrm{op}} \arrow[r, "\mathsf{Vect}_{\mathbf{k}}"] \arrow[rr, bend left, "K^0"] \arrow[d, "C_0"'] & \mathsf{Cat}^{\mathrm{add}} \arrow[r, "GC"] & \mathsf{Ab} \\ \mathsf{Rng} \arrow[ur, "\mathsf{Proj}"'] \arrow[urr, bend right, "K_0"'] & & \end{tikzcd}$$

where $\mathsf{Cat}^{\mathrm{add}}$ *denotes the subcategory of* Cat *consisting of additive categories and additive functors, and* $GC : \mathsf{Cat}^{\mathrm{add}} \to \mathsf{Ab}$ *denotes the Grothendieck Completion of an additive category.*

3.4. **Relating operator K-theory and algebraic K-theory.** The connection between operator K-theory and algebraic K-theory is very direct and thus can be stated concisely as follows: given a C^*-algebra A, the operator K_0 functor, which from now on we denote by $K_0^{\mathsf{C}^*}$ to avoid ambiguity, factors through the category Rng

along the algebraic K_0 functor K_0^{Rng}. Explicitly, we have the following commutative diagram of functors:

$$\begin{tikzcd} \mathsf{C^*Alg} \arrow[rr, "K_0^{\mathsf{C^*}}"] \arrow[dr, "U"'] & & \mathsf{Ab} \\ & \mathsf{Rng} \arrow[ur, "K_0^{\mathsf{Rng}}"'] & \end{tikzcd},$$

where U is the usual forgetful functor which forgets the $*$-operation and topology on a C^*-algebra.

We will establish the commutativity of the above diagram in two steps:

1. First we will outline a correspondence, natural in A, between the semigroup of $\mathsf{Proj}(U(A))$ of isomorphism classes of projective modules over ring $U(A)$ and algebraic equivalence classes (e.g. think the generalization of Murray-von Neumann equivalence to the Banach-algebra setting) of idempotents $\mathcal{I}_\infty(A)$ in the ring $\varinjlim_n M_n(A)$. This quotient set $\mathcal{I}_\infty(A)/\sim$ of algebraic equivalence classes then becomes a semiring under direct sum of matrices.
2. We will then show that the semigroup $\mathcal{I}_\infty(A)/\sim$ is isomorphic, via the semigroup homomorphism induced by the identity map 1_A, to the familiar semigroup $\mathcal{P}_\infty(A)/\sim$ of Murray-von Neumann equivalence classes in $\varinjlim_n M_n(A)$.

Definition 3.29. An **idempotent** in R is an element $e \in R$ such that $e^2 = e$. We denote the set of idempotents of R by $\mathcal{I}(R)$.

There is a close relationship between finitely generated projective R-modules and idempotents of the matrix ring $M_n(R)$ for various n.

Proposition 3.30. *An R-module P is projective iff there exists an idempotent element e in $M_n(R)$ such that $P \cong e(R^n)$, where $er \in R^n$, for $r \in R^n$, is defined by matrix multiplication.*

Proof. ($\Longrightarrow$). Since P is projective there exists an R-module Q such that $P \oplus Q \cong R^n$. Let $f : P \oplus Q \to R^n$ be an isomorphism and define $T : R^n \to R^n$ to be the projection map onto $f(P)$. Since T is the projection map onto a subspace, it is clearly idempotent. Furthermore, identifying $\mathrm{End}_R(R^n)$ with $M_n(R)$, the restriction of f to P gives a module isomorphism $P \cong f(P) \cong T(R^n)$.

($\Longleftarrow$). We have

$$R^n \cong \ker(e^2 - e) \cong \ker(e) \oplus \ker(e-1) \cong \ker(e) \oplus e(R^n) \cong \ker(e) \oplus P,$$

hence P is free. □

Idempotents in $M_n(R)$ can be thought of as idempotents in the infinite matrix ring $M_\infty(R) \coloneqq \varinjlim M_n(R)$ under the ring monomorphisms

$$i : M_n(R) \ni a \longmapsto \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} \in M_{n+1}(R).$$

The identification between projective R-modules and idempotents of the infinite matrix ring $M(R) \coloneqq \varinjlim_n M_n(R)$ associated to R given above is in fact not a 1-1 correspondence. In general, there are several choices of idempotents e in $M(R)$ which generate the same projective module $e(R^n)$. Such idempotents can in fact be characterized by a relatively simple algebraic relation.

Definition 3.31. Given idempotents e, f in a ring R, we say that e and f are **algebraically equivalent**, denoted $e \sim_a f$, if there exist $x, y \in R$ such that $e = xy$ and $f = yx$.

Proposition 3.32. *Suppose $e, f \in M(R)$ are idempotents. Then e and f are algebraically equivalent in $M(R)$ (under the natural embeddings) iff $P = e(R^n)$ and $Q = f(R^m)$ are isomorphic R-modules.*

As in the case of the set of projections of a C^*-algebra, there are other, stronger notions of equivalence for the set of idempotents on a ring.

Definition 3.33. Let $e, f \in \mathcal{I}(R)$.

1. We say e and f are **similar**, denoted $e \sim_s f$, if there exists $z \in R$ such that $f = zez^{-1}$.
2. We say e and q are homotopic, denote $e \sim_{h'} f$, if there exists a continuous path of idempotents in A from e to f.

Again these two equivalence relations, along with algebraic equivalence $\sim_a$ of idempotents, all agree in the matrix ring setting $M_n(R)$.

Repeating the constructions carried out in defining the operator K_0-group in section 2.1, we can conclude the following.

Proposition 3.34. *The set of algebraic equivalence classes of the direct limit $\mathcal{I}_\infty(R) = \varinjlim_n \mathcal{I}(M_n(R))$ is a semigroup under matrix addition.*

As in the case of projections on a C^*-algebra, this semigroup construction is functorial.

Proposition 3.35. *Given a homomorphism $f : R \to S$ of rings, the map*

$$f_* : \mathcal{I}_\infty(R)/\sim_a \longrightarrow \mathcal{I}_\infty(S)/\sim_a$$

induced by the ring homomorphism $M_\infty(R) \to M_\infty(S)$ given by applying f entry-wise is a semigroup homomorphism. Thus the mapping $R \mapsto \mathcal{I}_\infty(R)/\sim_a$ extends to a functor

$$\mathcal{I}^a_\infty : \mathsf{Rng} \longrightarrow \mathsf{SGrp}.$$

As stated in the beginning of this section, the next step in establishing the connection between operator K-theory and algebraic K-theory is proving that the semigroups $\mathcal{I}_\infty(A)/\sim_a$ and $\mathcal{P}_\infty(A)/\sim$ are isomorphic for a unital C^*-algebra A. This is done via the following three results.

Lemma 3.36. *For every idempotent element e in A, there exists a projection p in A such that $e \sim_a p$.*

On a C^*-algebra A, algebraic equivalence on projections coincides with Murray von Neumann equivalence.

Lemma 3.37. *Given projections p and q in A, $p \sim q$ iff $p \sim_a q$.*

Finally, we are ready to prove the existence of a natural isomorphism between $\mathcal{I}^{\sim_a}_\infty$ and $\mathcal{P}^{\sim}_\infty$ as functors on $\mathsf{C^*Alg}_1$.

Theorem 3.38. *The map*

$$f_A : \mathcal{P}_\infty(A)/\sim \longrightarrow \mathcal{I}_\infty(A)/\sim_a$$
$$[p]_\sim \longmapsto [p]_{\sim_a}$$

is a semigroup isomorphism that is natural in A.

This allows us to conclude the existence of the commutative diagram mentioned at the beginning of this section.

Theorem 3.39. *The following diagram of functors commutes:*

$$\begin{tikzcd} \mathsf{C^*Alg} \arrow[rr, "K_0^{\mathsf{C^*}}"] \arrow[dr, "U"'] & & \mathsf{Ab} \\ & \mathsf{Rng}. \arrow[ur, "K_0^{\mathsf{Rng}}"'] & \end{tikzcd}$$

Consequently, we can think of operator K-theory as a special case of algebraic K-theory. Indeed by the above theorem, the operator K-group $K_0(A)$ of a C^*-algebra A only requires the algebraic data of the ring-structure on A, from which we can derive the algebraic equivalence classes of idempotents in $\varinjlim M_\infty(A)$, hence simultaneously the Murray-von Neumann equivalence classes of projections in $\varinjlim M_\infty(A)$ and isomorphism classes of projective modules over A.

3.5. **Tying everything together.** We now have developed the machinery necessary to establish a K-theoretic braid which passes through the lands of algebraic topology, module theory, and operator algebras. Let us recall the connections we have made so far:

- Commutative Gelfand-Naimark 3.10 establishes that the category CHaus of compact Hausdorff spaces is dually equivalent to the category $\mathsf{CC^*Alg}_1$ of unital commutative C^*-algebras via the pair of functors

$$\mathsf{CHaus} \underset{\widetilde{(-)}}{\overset{C_0}{\rightleftarrows}} \mathsf{CC^*Alg}\ .$$

- The Serre-Swan theorem 3.17 establishes that the category $\mathsf{Vect}_{\mathbf{k}}(X)$ of $\mathbf{k}$-vector bundles over a compact Hausdorff space X is equivalent to the category $\mathsf{Proj}(C(X,\mathbf{k}))$, and furthermore, by Corollary 3.28, we have the following commutative diagram of functors:

$$\begin{tikzcd} \mathsf{Top}^{\mathrm{op}} \arrow[r, "\mathsf{Vect}_{\mathbf{k}}"] \arrow[rr, bend left, "K^0"] \arrow[d, "C_0"'] & \mathsf{Cat}^{\mathrm{add}} \arrow[r, "GC"] & \mathsf{Ab} \\ \mathsf{Rng} \arrow[ur, "\mathsf{Proj}"'] \arrow[urr, bend right, "K_0"'] & & \end{tikzcd}\ .$$

- Theorem 3.39 establishes that the operator K_0 functor $K_0^{\mathsf{C^*}}$ factors through the category Rng along the algebraic K_0 functor K_0^{Rng}. This is captured by the following commutative diagram of functors:

$$\begin{tikzcd} \mathsf{C^*Alg} \arrow[rr, "K_0^{\mathsf{C^*}}"] \arrow[dr, "U"'] & & \mathsf{Ab} \\ & \mathsf{Rng}. \arrow[ur, "K_0^{\mathsf{Rng}}"'] & \end{tikzcd}$$

We are now ready to prove Theorem 3.13, completing the final chain in our mathematical link connecting the various notions of K-theory.

Proof of Theorem 3.13. By Lemma 3.12, it suffices to prove the case for $n = 0$. This follows immediately from the following commutative diagram of functors, given by combining the results outlined in 3.5:

$$\begin{tikzcd}
\mathsf{Top}^{\mathrm{op}} \arrow[rr, "C_0", "\simeq"'] \arrow[d, "\mathsf{Vect}_{\mathbb{C}}"] \arrow[ddr, bend right, "K^0"'] & & \mathsf{CC^*Alg}_1 \arrow[dl, "U"'] \arrow[d, "\mathcal{P}^{\sim}_{\infty}"'] \arrow[ddl, bend left, "K_0^{\mathsf{C}^*}"] \\
\mathsf{Cat}^{\mathrm{add}} \arrow[ddr, "GC"'] & \mathsf{Rng} \arrow[l, "\mathsf{Proj}"] \arrow[r, "\mathcal{I}^{\sim a}_{\infty}"'] \arrow[dd, "K_0^{\mathsf{Rng}}"] & \mathsf{SGrp} \arrow[ddl, "UEG"] \\
& & \\
& \mathsf{Ab} &
\end{tikzcd}$$

In particular, we have commutativity of the arrows in blue, corresponding to the fact that, up to natural isomorphism, the K^0 cohomology functor factors through the category $\mathsf{CC^*Alg}_1$ of commutative unital C^*-algebras, along the operator K-theory functor $K_0^{\mathsf{C}^*}$, as required. □

The commutative diagram of functors in the above proof illustrates the K-theoretic thread we have woven throughout the mathematical landscape in its entirety.

4. Conclusion

We have demonstrated the existence of clear formal connections tying together the notions of K-theory in the settings of operator algebras, algebraic topology, and algebraic geometry. With these relevant facts established, it seems almost indisputable that the various constructions of the K-group outlined above (whether they be for C^*-algebras, topological spaces, or groups) share enough formal similarities to justify being grouped together under the common heading of "K-theory".

Beginning with Alexander Grothendieck's foray into algebraic varieties in the late 50s, the development of K-theory has come a long way to comprise the wide array of theorems and constructions which we have outlined in this paper. In all the contexts in which it is used, K-theory has and continues to be both an extremely powerful and flexible algebraic invariant. It has been leveraged to prove groundbreaking theorems in all three of the settings of algebraic geometry, algebraic topology, and operator algebras. These include

- Grothendieck's formulation and proof of the Grothendieck-Riemann-Roch theorem [2];
- Atiyah and Isadore Singer's original proof of their renowned index theorem [1];
- George Elliott's complete classification of AF-algebras. [3]

The proof of any one of the these results would have more than justified considering K-theory to be an indispensable tool in the mathematician's toolkit. The fact that all three of the above results rely on K-theory, along with a vast range of other developments in the fields of algebraic geometry, algebraic topology, and operator algebras, solidifies K-theory as being one of the most signifiant mathematical developments of the twentieth century. In a world of ever-increasing mathematical

specialization, K-theory stands as one of the few remaining areas of active mathematical research which serves as common intellectual ground on which analysts, algebraists, and topologists can convene.

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